Forecasting Next-Week Volatility of S&P 500 Stocks: Revised Proposal

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1. Project question

For each S&P 500 stock and week, we forecast next week's log realized variance, measured by the Garman–Klass estimator from daily prices [9]. Models train on 2005–2015, tune on 2016–2018 and are tested on 2019–2026 with yearly expanding refits, a one-week gap at each boundary and no tuning on test data.

1. **Accuracy.** Do pooled models fit on all stocks (ridge and lasso, then gradient-boosted trees) beat a HAR regression fit per stock [6] on mean QLIKE [11] and MSE? The Diebold–Mariano test [7] runs at 5% on the weekly cross-sectional mean loss difference with Newey–West errors. Per-stock HAR needs 52 weeks of history (younger stocks use a pooled HAR), and every model is scored on the same rows.
2. **Attribution and regimes.** Which feature blocks carry any gain (refitting without each block), and does the gain keep its sign in calm weeks and in stress weeks (VIX ≥ 30 on the Friday)?

It is answered when, for each model, the test-period table of losses and Diebold–Mariano statistics against HAR, and the block-ablation table split by regime, exist.

How and why it changed. The economic-value question (volatility-targeted portfolio, Sharpe ratio, drawdown, Fama–French regression) is dropped, as the staff notes require. Earnings timing comes from the filing times of SEC 8-K Item 2.02 releases, a proxy for the announcement, not quarterly filings that arrive weeks after it. Fundamentals from XBRL are deferred: they start in mid-2009 and the question does not need them. Stress is defined by VIX at prediction time, not by calendar years, and qlib's Alpha158 price-volume factors [13] join the features.

2. Data

Four sources from the original proposal: S&P 500 membership by date [3] (2,720 snapshots, 977 tickers since 2005, 950 companies once renames are merged); market state from FRED [8] (VIX, 3-month VIX, credit spread, curve slope, T-bill); daily prices from Yahoo Finance [2] (677 tickers, 2005–2026); earnings-release filing times and industry codes from SEC EDGAR [12]. Two added since: Alpaca [1], whose daily bars from 2016 fill 165 members Yahoo lacks and whose 5-minute bars check the target, and CRSP [4], every member's daily prices to 2025 through WRDS, for the survivorship check. The raw table has one row per stock and trading day, 3,423,011 rows (473 MB), 2,315,615 after cleaning. The modeling table has **473,684 rows and 183 columns**: one row per stock and week (725 companies, 1,139 weeks), holding the target and every feature known at that Friday's close.

3. What we learned from the data

- **Survivorship.** Yahoo drops delisted companies. We recovered 30 renamed tickers (BK became BNY) and filled members Yahoo lacks from Alpaca from 2016; 108 tickers, all gone before 2016, have no free data. Coverage of index members is 55% in 2005, 99.4% in 2019 and 99.8% in 2026, so pre-2016 training data leans toward survivors. A check against CRSP (WRDS, every member) measures the cost: from 2005 to 2015 the free table has 191,083 stock-weeks against 286,597, and its mean log variance is 0.04 below CRSP's in the same year, on average (about 4% less variance); from 2016 the gap is under 0.01, and from 2019 under 1% of stock-weeks are missing. We propose to train on CRSP to 2025.
- **Ticker reuse and wrong series.** Yahoo reuses symbols (today's `FB` is an ETF), so prices are kept only near each ticker's index spells; 9 series inside a spell belong to other securities (Yahoo's `PARA` trades near \$100,000), and 4 change company on a date (Yahoo's `JCI` before 2016 is Tyco); hand-checked tables remove both.
- **The target is a sound proxy.** Against 5-minute realized variance for 697 companies since 2016, delisted ones included (281,250 stock-weeks), log weekly Garman–Klass variance has correlation 0.95 with it, and 0.94 within company, easing the proposal's worry about noisy daily targets; weeks before 2016 are not covered by this check.

- **Stress is rare.** 7.3% of stock-weeks are stress weeks, so results must be reported per regime, not as one pooled loss.
- **Leakage.** Feature windows end at the Friday close (unit tests truncate the data and recompute), an 8-K filed after 16:00 counts for the next week, membership is point-in-time (the last snapshot, 2026-08-18, is carried forward), and every price feature is a ratio, so back-adjusted prices cannot leak.

4. Plan for the rest of the project

Features: HAR lags (day, week, four weeks); Parkinson, Rogers–Satchell, overnight and downside variance; returns; dollar volume and Amihud illiquidity; SPY variance; VIX, 3-month VIX, credit spread, curve slope, T-bill; weeks since earnings, an expected-earnings flag and industry; Alpha158. **Models**: per-stock HAR and GARCH(1,1) baselines; pooled ridge and lasso on standardized features with training-window imputation; gradient-boosted trees (LightGBM). Clustering and neural networks only if time remains [5, 10]. **Evaluation**: QLIKE and MSE on 2019–2026, Diebold–Mariano tests against HAR, block ablation, every result split into calm and stress weeks.

5. Staff notes and the AC209A requirement

Economic-value backtest: dropped. Regularized regression and boosted trees: the two pooled models of Q1; clustering and neural networks wait. yfinance failures and missing delisted stocks: fetches are cached and resumable (the notebook runs with no keys from a team copy), renames, wrong series and the Alpaca fill are documented, and coverage is reported. XBRL only from mid-2009: fundamentals deferred. AC209A outside method: the Diebold–Mariano test on QLIKE. HAR and GARCH(1,1) are the baselines; none of these is covered in class.

Use of generative AI. Claude Code (Anthropic) helped write the data pipeline, tests and this text; the group reviewed every step and verified the numbers, which the notebook computes and exports.

References

[1] Alpaca Markets. Market data API. `https://docs.alpaca.markets/docs/about-market-data-api`, 2026.

[2] Ran Aroussi. yfinance: Download market data from Yahoo! Finance's API. `https://github.com/ranaroussi/yfinance`, 2026. Python package.

[3] Farrell Aultman. sp500: Current and historical lists of S&P 500 components since 1996. `https://github.com/fja05680/sp500`, 2026. GitHub repository, commit a2430f2 of 2026-09-07.

[4] Center for Research in Security Prices. CRSP US stock database. Wharton Research Data Services, `https://wrds-www.wharton.upenn.edu`, 2026. Daily stock file and S&P 500 constituents to 2025-12-31.

[5] Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023.

[6] Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2): 174–196, 2009. doi: 10.1093/jjfinec/nbp001.

[7] Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13 (3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

[8] Federal Reserve Bank of St. Louis. FRED, Federal Reserve Economic Data. `https://fred.stlouisfed.org`, 2026.

[9] Mark B. Garman and Michael J. Klass. On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1):67–78, 1980. doi: 10.1086/296072.

[10] Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020. doi: 10.1093/rfs/hhaa009.

[11] Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1): 246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.

[12] U.S. Securities and Exchange Commission. EDGAR application programming interfaces. `https://www.sec.gov/search-filings/edgar-application-programming-interfaces`, 2026.

[13] Xiao Yang, Weiqing Liu, Dong Zhou, Jiang Bian, and Tie-Yan Liu. Qlib: An AI-oriented quantitative investment platform, 2020.